United States Patent Application Publication	20250285084
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	Washida; Kazuhisa

Reservation Assistance Device, Reservation Assistance Method, and Program

Abstract

A reservation assistance device includes: an obtaining unit configured to store status data indicating whether or not a reservation from a customer is acceptable for each of unit time periods and obtain a requested start time and a requested end time of the reservation requested by the customer; and a proposal unit that refers to the status data, and when the reservation is acceptable in a period from the requested start time to the requested end time and a difference time period between the requested start time and an end time of a reservation unacceptable time period immediately before the requested start time is equal to or larger than the unit time period and is less than a predetermined time period, outputs a proposal of changing the requested start time to the end time of the reservation unacceptable time period immediately before the requested start time, to the customer.

Inventors:	**Washida; Kazuhisa (Toshima-ku, Tokyo, JP)**
Applicant:	**COMING SOON CoO., Ltd.** (Toshima-ku, Tokyo, JP)
Family ID:	**88518141**
Appl. No.:	**18/860351**
Filed (or PCT Filed):	**April 28, 2023**
PCT No.:	**PCT/JP2023/016904**

Foreign Application Priority Data

WO	PCT/JP2022/019252	Apr. 28, 2022

Publication Classification

Int. Cl.: **G06Q10/1093** (20230101)

U.S. Cl.:

CPC **G06Q10/1095** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a National Stage application under 35 U.S.C. § 371 of International Patent Application Serial No. PCT/JP2023/016904, filed Apr. 28, 2023, which claims priority to International Patent Application Serial No. PCT/JP2022/019252, filed Apr. 28, 2022, the entire disclosures of which is hereby incorporated by reference.

TECHNICAL FIELD
[0002] The present invention relates to a reservation assistance device, a reservation assistance method, and a program.
BACKGROUND
[0003] There is a device that optimizes a reservation schedule in a salon having multiple seats such as a beauty salon, a massage, or a nail salon (see Japanese Patent Application Publication No. 2007-156831). In Japanese Patent Application Publication No. 2007-156831, reservations whose reservation times are close to one another are grouped, and are rearranged in each group. Japanese Patent Application Publication No. 2007-156831 can optimize the reservations, and enables accepting of more reservations.
SUMMARY
[0004] However, in Japanese Patent Application Publication No. 2007-156831, an in-between time period in which a new reservation is unacceptable occurs. Reducing the in-between time period enables accepting of more reservations.
[0005] The present invention has been made in view of the above circumstances, and an object of the present invention is to provide a technique that enables accepting of more reservations.
[0006] A reservation assistance device according to an aspect of the present invention includes: a storage device configured to store status data indicating whether or not a reservation from a customer is acceptable for each of unit time periods; an obtaining unit configured to obtain a requested start time and a requested end time of the reservation requested by the customer; and a proposal unit configured to refer to the status data, and when the reservation is acceptable in a period from the requested start time to the requested end time and a difference time period between the requested start time and an end time of a reservation unacceptable time period immediately before the requested start time is equal to or larger than the unit time period and is less than a predetermined time period, output a proposal of changing the requested start time to the end time of the reservation unacceptable time period immediately before the requested start time, to the customer.
[0007] A reservation assistance device according to another aspect of the present invention includes: a storage device configured to store status data indicating whether or not a reservation from a customer is acceptable for each of unit time periods; and a proposal unit configured to refer to the status data, and when a difference time period between a first time period in which the reservation is unacceptable and a second time period in which the reservation is unacceptable and that is immediately after the first time period is equal to or larger than the unit time period and is less than a predetermined time period, propose a change of the first time period or the second time period.
[0008] A reservation assistance device according to still another of the present invention includes: a storage device configured to store status data indicating whether or not a reservation from a

customer is acceptable for each of unit time periods; an obtaining unit configured to obtain a requested time period of the reservation requested by the customer; and an identification unit configured to refer to the status data and identify, from among start candidate times at which the reservation of the requested time period is acceptable, the start candidate time at which a difference time period between the start candidate time and an end time of a reservation unacceptable time period immediately before the start candidate time is zero or equal to or more than a predetermined time period or a difference time period between a time at which the requested time period has elapsed from the start candidate time and a start time of the reservation unacceptable time period immediately after the time is zero or equal to or more than the predetermined time period; and a display unit configured to display the start candidate time identified by the identification unit, to the customer.

[0009] A reservation assistance method according to an aspect of the present invention includes: a computer stores status data indicating whether or not a reservation from a customer is acceptable for each of unit time periods; the computer obtains a requested start time and a requested end time of the reservation requested by the customer; and the computer refers to the status data, and when the reservation is acceptable in a period from the requested start time to the requested end time and a difference time period between the requested start time and an end time of a reservation unacceptable time period immediately before the requested start time is equal to or larger than the unit time period and is less than a predetermined time period, outputs a proposal of changing the requested start time to the end time of the reservation unacceptable time period immediately before the requested start time.

[0010] A reservation assistance method according to another aspect of the present invention includes: a computer stores status data indicating whether or not a reservation from a customer is acceptable for each of unit time periods; and the computer refers to the status data, and when a difference time period between a first time period in which the reservation is unacceptable and a second time period in which the reservation is unacceptable and that is immediately after the first time period is equal to or larger than the unit time period and is less than a predetermined time period, proposes a change of the first time period or the second time period.

[0011] A reservation assistance method according to still another aspect of the present invention includes: a computer stores status data indicating whether or not a reservation from a customer is acceptable for each of unit time periods; the computer obtains a requested time period of the reservation requested by the customer; and the computer refers to the status data and identifies, from among start candidate times at which the reservation of the requested time period is acceptable, the start candidate time at which a difference time period between the start candidate time and an end time of a reservation unacceptable time period immediately before the start candidate time is zero or equal to or more than a predetermined time period and a difference time period between a time at which the requested time period has elapsed from the start candidate time and a start time of the reservation unacceptable time period immediately after the time is zero or equal to or more than the predetermined time period; and causing the computer to display the identified start candidate time to the customer.

[0012] An aspect of the present invention is a program that causes a computer to function as the above reservation assistance.

[0013] The present invention can provide a technique that enables accepting of more reservations.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. **1** is a diagram explaining a system configuration of a processing system and function blocks of a reservation assistance device according to an embodiment of the present invention;

[0015] FIG. **2** is an example of a status display screen displayed by the reservation assistance device;
[0016] FIG. **3** is an example of a presentation screen displayed by the reservation assistance device;
[0017] FIG. **4** is a diagram explaining an example of a data structure and data of status data;
[0018] FIG. **5** is an example of a setting screen in which information included in setting data is set;
[0019] FIG. **6** is a diagram explaining an example of a data structure and data of reservation data;
[0020] FIG. **7** is a diagram explaining an example of a data structure and data of staff data;
[0021] FIG. **8** is a flowchart explaining processing by the reservation assistance device;
[0022] FIG. **9** is a flowchart explaining proposal processing by a proposal unit according to the embodiment of the present invention;
[0023] FIG. **10** is a diagram explaining the system configuration of the processing system and the function blocks of the reservation assistance device according to a first modified example;
[0024] FIG. **11** is a flowchart explaining the proposal processing by the proposal unit according to the first modified example;
[0025] FIG. **12** is a diagram explaining the system configuration of the processing system and the function blocks of the reservation assistance device according to a second modified example;
[0026] FIG. **13** is an example of a selection screen displayed by the proposal unit according to the second modified example;
[0027] FIG. **14** is a flowchart explaining the proposal processing by the proposal unit according to the second modified example;
[0028] FIG. **15** is an example of a screen in which a customer inputs setting relating to allowance of reservation change in a third modified example;
[0029] FIGS. **16**A and B are examples of a screen in which sub-start candidate times are displayed or hidden in a fourth modified example;
[0030] FIG. **17** is a flowchart explaining processing of the reservation assistance device in the fourth modified example (part **1**);
[0031] FIG. **18** is a flowchart explaining the processing of the reservation assistance device in the fourth modified example (part **2**); and
[0032] FIG. **19** is a diagram explaining a hardware configuration of a computer used as the reservation assistance device.

DETAILED DESCRIPTION

[0033] An embodiment of the present invention is explained below with reference to the drawings. The same parts in illustration of the drawings are denoted by the same reference numerals, and explanation thereof is omitted.
[0034] A reservation in the embodiment of the present invention is an appointment of providing a service from a reservation start time to a reservation end time that is agreed between a customer and a staff. The reservation secures resources such as a place, a staff, and the like for the customer in a shop such as a beauty salon, a nail salon, a massage salon, a dental clinic, a restaurant, a karaoke box, or a bowling alley. Although the place where the reservation is provided is referred to as shop in the embodiment of the present invention, the place is not limited to the shop. A reservation assistance device **1** is not limited to use in the shop, and may be used in businesses in general in which reservations are accepted such as visiting-type businesses that provide repair services by visiting a place of the customer and on-line type businesses such as language lessons and call taker using the Internet.
[0035] In the embodiment of the present invention, the reservation start time and the reservation end time are a start time and an end time of the reservation agreed between the customer and the staff. A reservation time period is a duration of a time period from the reservation start time to the reservation end time, and the customer receives services from the staff within the reservation time period. The reservation start time, the reservation end time, and the reservation time period sometimes vary from a requested start time, a requested end time, and a requested time period

inputted by the customer at a moment of reservation, due to a proposal by the reservation assistance device **1** and acceptance of the proposal by the customer. A unit time period is a minimum unit of identification of times such as the reservation start time, the reservation end time, the reservation time period, the requested start time, the requested end time, and the requested time period in the reservation assistance device **1**.

[0036] In the embodiment of the present invention, when a difference time period between a first time period in which the reservation is unacceptable and a second time period in which the reservation is unacceptable and that is immediately after the first time period is equal to or more than the unit time period and is less than a predetermined time period for a predetermined reservation target such as a staff or a room, this difference time period is referred to as an in-between time period. The unit time period is a minimum time period of the in-between time period. The predetermined time period may be a minimum time period of the reservation time period, or may be a time period set in advance by the staff or the like. The minimum time period of the reservation time period may be a minimum time period secured in the reservation, and may be determined from, for example, a shortest time period in a menu provided by the shop. In the embodiment of the present invention, explanation is given of the case where the unit time period is 15 minutes and the predetermined time period (minimum time period of the reservation time period) is one hour.

Processing System

[0037] A processing system **5** enables accepting of many reservations by proposing a change of the requested start time or the requested end time of the reservation to the customer at the moment of reservation and eliminating the in-between time period. The processing system **5** includes the reservation assistance device **1**, a customer terminal **2**, and a staff terminal **3**. The reservation assistance device **1** can communicate with each of the customer terminal **2** and the staff terminal **3**.

[0038] General computers execute programs that perform predetermined processing to implement the reservation assistance device **1**, the customer terminal **2**, the staff terminal **3**. The customer who is the subject of the reservation uses the customer terminal **2**. The staff who provides the service requested by the reservation uses the staff terminal **3**. The reservation assistance device **1** checks requested reservation data **11** inputted by the customer terminal **2**, and proposes a change of the reservation to the customer terminal **2** such that no in-between time period occurs.

[0039] The reservation is identified based on, for example, the reservation start time, the reservation end time, and the reservation time period from the reservation start time to the reservation end time agreed with the customer. Each staff generally accepts multiple reservations in one day. When the in-between time period occurs between reservations, the number of reservations that the staff can accept in one day is sometimes is reduced from that in the case where no in-between time period occurs. Accordingly, the reservation assistance device **1** proposes a change of the reservation to the customer in acceptance of the reservation to reduce the in-between time period and assist streamlining of businesses.

[0040] The in-between time period is explained with reference to FIG. **2**. FIG. **2** is a status display screen V**1** that displays a reservation status of a staff K in a beauty parlor and a work status of the staff K in a timeline. The unit time period in the example illustrated in FIG. **2** is 15 minutes, and the reservation start time, the reservation end time, and the reservation time period are designated in units of 15 minutes. In FIG. **2**, the staff K plans to accept a reservation P**101** from 9:30 to 10:30 and a reservation P**103** from 16:00 to 18:30 and take a break P**102** from 13:00 to 14:00. In the example of FIG. **2**, reservation unacceptable time periods are identified based on the reservations P**101** and P**103** and the break P**102**.

[0041] Assume a case where a reservation P**104** of one hour from 14:30 is newly made at a request of the customer in such a situation. Assuming that the reservation P**104** is accepted as requested by the customer, a period from 14:00 to 14:30 and a period from 15:30 to 16:00 are each 30 minutes. When there is no practice menu that can be provided in 30 minutes, the period from 14:00 to 14:30

and the period from 15:30 to 16:00 become the in-between time periods in which no new reservation is unacceptable.

[0042] Accordingly, the reservation assistance device **1** displays a proposal screen V**2** illustrated in FIG. **3** for the customer terminal **2** of the customer who is attempting to newly make the reservation of one hour from 14:30. The proposal screen V**2** is a screen that proposes a change of reservation, and includes a change button B**201** that changes the reservation start time from 14:30 to 14:00, a change button B**202** that changes the reservation start time from 14:30 to 15:00, and a no-change button B**203** that leaves the reservation start time at 14:30. The customer selects the change button B**201**, the change button B**202**, or the no-change button B**203** according to the convenience of the customer himself/herself on the screen illustrated in FIG. **3**.

[0043] When the customer accepts the change of the reservation start time to 14:00 and selects the change button B**201** in the customer terminal **2**, the reservation assistance device **1** changes the reservation start time from 14:30 to 14:00. In this case, a period from 15:00 to 16:00 is a free time period with no reservation, and a free time period in which a new reservation of one hour is acceptable is created. Similarly, when the customer accepts the change of the reservation start time to 15:00 and selects the change button B**202** in the customer terminal **2**, the reservation assistance device **1** sets the reservation start time from 14:30 to 15:00. In this case, a period from 14:00 to 15:00 becomes a free time period with no reservation, and a free time period in which a new reservation of one hour is acceptable is created.

[0044] The reservation assistance device **1** can assist a profit increase by eliminating the in-between time period and creating a continuous free time period in which a new reservation is acceptable or by proposing an additional practice. Moreover, the reservation assistance device **1** proposes a discount such as a coupon to the customer with the change in reservation as described above. This makes it more likely for the customer to accept the proposal, and provides benefits to both of the customer and the shop.

[0045] As illustrated in FIG. **1**, the reservation assistance device **1** includes pieces of data of the requested reservation data **11**, status data **12**, setting data **13**, new reservation data **14**, reservation data **15**, and staff data **16** and functions of an obtaining unit **21**, a proposal unit **22**, and an update unit **23**. The pieces of data are stored in a storage device such as a memory **902** or a storage **903**. The functions are implemented by a CPU **901**. The data structure of each piece of data and the flowchart explaining the processing of each function according to the embodiment of the present invention are examples, and the data structure and the flowchart are not limited to these.

[0046] The requested reservation data **11** is data for identifying the reservation inputted from the customer terminal **2** and requested by the customer. The requested reservation data **11** includes a customer name and the requested start time, the requested end time, and the requested time period of the reservation.

[0047] The status data **12** is data indicating whether or not the reservation from the customer is acceptable for each of the unit time periods. As illustrated in FIG. **4**, in the status data **12**, information on acceptation or non-acceptation of reservation such as outside the business hours, the reservation agreed with the customer, and the break of the staff is set for each group of unit time periods in which acceptation or non-acceptation of reservation continues.

[0048] For example, the status data **12** is generated by identifying the acceptation or non-acceptation of the reservation for each unit time period based on the reservation data **15** for identifying the reservation agreed with each customer, the staff data **16** for identifying the schedule of each staff such as working hours (shift) of the staff, break hours of the staff, and the like. The status data **12** may be generated by identifying the acceptation or non-acceptation of the reservation for each unit time period based on usage statuses of resources secured in providing of a service such as a room, a seat, a table, and a tool, the number of customers for which the staff can simultaneously perform practice, and the like. Although the status data **12** illustrated in FIG. **4** is data relating to the staff K, the status data **12** can be generated for each staff as well as for each

reservation target of the customer such as each table and each room.
[0049] Although the case where the reservation assistance device **1** generates the status data **12** in advance is explained in the embodiment of the present invention, the present invention is not limited to this. The status data **12** may be data in which the acceptation or non-acceptation of the new reservation for each unit time period is dynamically identified based on the reservation data **15**, the staff data **16**, and the like at a timing at which the reservation assistance device **1** refers to the status data **12**.
[0050] The setting data **13** includes a determination condition to which the reservation assistance device **1** refers to determine the in-between time period, a proposal message to which the reservation assistance device **1** refers when proposing the change of the reservation time period to the customer terminal **2**, and the like. The setting data **13** may be set by default, or generated according to information set by a person in charge of the shop or the like, in a setting screen V**3** illustrated in FIG. **5**.
[0051] The setting screen V**3** includes an exclusion control portion P**301**, an in-between time period definition portion P**302**, a title setting portion P**303**, and a message setting portion P**304**. The reservation assistance device **1** refers to conditions whose check boxes are checked in the exclusion control portion P**301** and the in-between time period definition portion P**302** in determination of the in-between time period, as effective conditions.
[0052] A time zone to be excluded as a target of the determination of the in-between time period is set in the exclusion control portion P**301**. In the exclusion control portion P**301** of FIG. **5**, a time period less than 60 minutes from a shift start time is set not to be determined as the in-between time period.
[0053] An upper limit of a time period to be the in-between time period is set in the in-between time period definition portion P**302**. Explanation is given of the case where a time period in which there is no reservation and that is the unit time period or more and 60 minutes of less is set as the in-between time period in the in-between time period definition portion P**302** of FIG. **5**. In the in-between time period definition portion P**302** of FIG. **5**, a check box “do not display a reservation free time period that causes the in-between time period” is provided. When this check box is checked, only the times for which the in-between time period does not occur are displayed as candidates of the reservation start time presented to the customers as described in a second modified example to be described later.
[0054] Templates of proposal messages proposing the change of the reservation time period from the reservation assistance device **1** to the customer terminal **2** are set in the title setting portion P**303** and the message setting portion P**304**. Information set in the title setting portion P**303** and the message setting portion P**304** is displayed in the proposal screen V**2** as illustrated in FIG. **3**.
[0055] Although explanation is given of the case where the setting data **13** is generated based on information inputted in the setting screen V**3**, the present invention is not limited to this. For example, the upper limit of the in-between time period that can be set in the in-between time period definition portion P**302** may be set to a providing time period of a service that is provided at the minimum time period in the shop or may be set by default. Each piece of information in the title setting portion P**303** and the message setting portion P**304** may be set by default. Moreover, settings other than those illustrated in FIG. **5** may be set in the setting data **13**.
[0056] The new reservation data **14** is data for identifying the reservation agreed with the customer. The new reservation data **14** in the embodiment of the present invention is generated by the proposal unit **22**. The new reservation data **14** includes the customer name, the reservation start time, the reservation end time, and the reservation time period.
[0057] The reservation data **15** is data for identifying the start time and the end time of the reservation accepted from the customer. As illustrated in FIG. **6**, the reservation data **15** holds the time period of the reservation, the name of the staff responsible for the reservation, the name of the reserved customer, and the reserved practice contents in association with one another. The

reservation data **15** may also be used to identify a method of contacting the customer such as a mail address of the customer.

[0058] The staff data **16** is data for identifying the shift hours and the break hours of each staff. The staff data **16** may also be used to identify a method of contacting a staff such as a mail address of the staff.

[0059] The obtaining unit **21** obtains the requested start time, the requested end time, and the requested time period of the reservation requested by the customer. When the obtaining unit **21** obtains the requested start time and the requested practice contents of the reservation from the customer, the obtaining unit **21** may identify the requested time period based on the practice contents, and identify the requested end time based on the requested start time and the requested time period. The obtaining unit **21** outputs the requested reservation data **11** including the requested start time, the requested end time, and the requested time period.

[0060] The proposal unit **22** refers to the status data **12**, and checks whether the reservation is acceptable in the period between the requested start time and the requested end time. When the period between the requested start time and the requested end time includes the reservation unacceptable time period in the status data **12**, the proposal unit **22** sends back a reply indicating that the reservation is unacceptable. When the reservation is acceptable in the period between the requested start time and the requested end time and the in-between time period occurs, the proposal unit **22** outputs a proposal for eliminating the in-between time period. In this case, if the time zone to be excluded as the target of the determination of the in-between time period is set in the exclusion control portion P**301** or the like of FIG. **5**, the proposal unit **22** determines presence or absence of the in-between time period in a time zone to be a target of the determination of the in-between time period.

[0061] First, the case where the in-between time period is present before the requested start time is explained. When the in-between time period occurs before the requested start time, the proposal unit **22** proposes advancing of the requested start time.

[0062] When the difference time period between the requested start time and the end time of the reservation unacceptable time period immediately before the requested start time is equal to or more than the unit time period and is less than the predetermined time period, the proposal unit **22** determines that the in-between time period is present before the requested start time. In this case, the proposal unit **22** outputs a proposal of changing the requested start time to the end time of the reservation unacceptable time period immediately before the requested start time, to the customer.

[0063] Explanation is given with reference to FIG. **2**. Explanation is given of the case where the reservation P**101**, the reservation P**103**, and the staff break P**102** are set as the reservation unacceptable time periods in the status data **12** and the customer requests the new reservation P**104**. The requested start time of the new reservation P**104** is 14:30, the requested end time is 15:30, and the requested time period is one hour. Moreover, the unit time period is 15 minutes. The minimum time period (predetermined time period) of the free time period that is a time period with no reservation and in which the reservation is acceptable is one hour, and the in-between time period is less than one hour (predetermined time period).

[0064] In the example illustrated in FIG. **2**, the proposal unit **22** determines that a period from 14:00 that is the end time of the reservation unacceptable time period immediately before the requested start time to 14:30 that is the requested start time, as the in-between time period. The proposal unit **22** displays a proposal screen that proposes a change of the requested start time from 14:30 to 14:00, on the customer terminal **2**. When an instruction of accepting the proposal is inputted from the customer terminal **2**, the proposal unit **22** generates the new reservation data **14** in which the reservation start time is set to 14:00 forwarded from the requested start time. The in-between time period before the requested start time can be thereby eliminated.

[0065] In this case, the proposal unit **22** sets the reservation end time of the new reservation data **14** to a time at which the requested time period has elapsed from the reservation start time.

Specifically, the proposal unit **22** sets the reservation end time to 15:00 that is a time at which the requested time period of one hour has elapsed from the reservation start time 14:00.
[0066] Alternatively, the proposal unit **22** may propose extension of the requested time period to the customer terminal **2** without changing the requested end time. When the instruction of accepting the proposal is inputted from the customer terminal **2**, the proposal unit **22** may set the reservation end time of the new reservation data **14** to the requested end time, and set the requested time period to a time period obtained by adding a difference time period (forwarding time period) between the requested start time and the reservation start time to the requested time period. For example, when the requested time is determined depending on practice contents of a beauty parlor or the like in the reservation, the proposal unit **22** proposes an additional practice such as a treatment as the extension of the requested time period, without changing the requested end time. When the instruction of accepting the proposal is inputted from the customer terminal **2**, the proposal unit **22** sets the reservation end time in the new reservation data **14** to the requested end time, sets the requested time period to an hour and a half obtained by adding the forwarding time period of 30 minutes to the requested time period of one hour, and generates the new reservation data **14**. The reservation assistance device I can thereby eliminate the in-between time period before the requested start time, and also receive the reservation of additional practice.
[0067] Next, explanation is given of the case where the in-between time period is present after the requested end time. When the in-between time period occurs after the requested end time, the proposal unit **22** proposes postponing of the requested end time.
[0068] When the difference time period between the requested end time and the start time of the reservation unacceptable time period immediately after the requested end time is equal to or more than the unit time period and is less than the predetermined time period, the proposal unit **22** determines that the in-between time period is present after the requested end time. In this case, the proposal unit **22** outputs a proposal of changing the requested end time to the start time of the reservation unacceptable time period immediately after the requested end time.
[0069] In the example illustrated in FIG. **2**, the proposal unit **22** determines a period from 15:30 that is the requested end time to 16:00 that is the start time of the reservation unacceptable time period immediately after the requested end time as the in-between time period. The proposal unit **22** displays a proposal screen that proposes a change of the requested end time from 15:30 to 16:00 on the customer terminal **2**. When the instruction of accepting the proposal is inputted from the customer terminal **2**, the proposal unit **22** generates the new reservation data **14** in which the reservation end time is set to 16:00 that is postponed from the requested end time. The in-between time period after the requested end time can be thereby eliminated.
[0070] In this case, the proposal unit **22** sets the reservation start time of the new reservation data **14** to a time the requested time period before the reservation end time. Specifically, the proposal unit **22** sets the reservation start time to 15:00 that is the requested time period of one hour before the reservation end time 16:00.
[0071] Alternatively, the proposal unit **22** may propose extension of the requested time period to the customer terminal **2** without changing the requested start time. When the instruction of accepting the proposal is inputted from the customer terminal **2**, the proposal unit **22** may set the reservation start time of the new reservation data **14** to the requested start time, and set the requested time period to a time period obtained by adding a difference time period (postponing time period) between the requested end time and the reservation end time to the requested time period. For example, when the requested time is determined depending on practice contents of a beauty parlor or the like in the reservation, the proposal unit **22** proposes an additional practice such as a treatment as the extension of the requested time period, without changing the requested start time. When the instruction of accepting the proposal is inputted from the customer terminal **2**, the proposal unit **22** sets the reservation start time in the new reservation data **14** to the requested end time, sets the requested time period to an hour and a half obtained by adding the postponing

time period of 30 minutes to the requested time period of one hour, and generates the new reservation data **14**. The reservation assistance device I can thereby eliminate the in-between time period after the requested start time, and also receive the reservation of additional practice.

[0072] When the proposal unit **22** determines that the in-between time periods are present in both of a period before the requested start time and a period after the requested end time in the requested reservation data **11**, the proposal unit **22** may present the proposal of advancing the requested start time and the proposal of postponing the requested end time to the customer as in FIG. **3**. Moreover, the proposal unit **22** may present the proposal of extending the requested time period to the customer in addition to the proposals illustrated in FIG. **3**.

[0073] Although explanation is given of the case where the proposal unit **22** displays the proposal screen V**2** in which the start time of the reservation is selected such that the in-between time period is eliminated, the proposal unit **22** may display choices of the start time of the reservation that can reduce the in-between time period. For example, in the example illustrated in FIG. **3**, a choice of changing the start time to 14:15 that is shifted by the unit time period within a time period in which the in-between time period occurs and a choice of changing the start to 14:45 may be displayed. The in-between time period cannot be eliminated but can be reduced, and a new reservation can be received in some cases.

[0074] When the proposal unit **22** generates the new reservation data **14** according to the reply of the customer, the proposal unit **22** inputs the generated new reservation data **14** into the update unit **23**. The update unit **23** updates the reservation data **15** and the status data **12** according to the new reservation data **14**.

Reservation Assistance Method

[0075] A reservation assistance method according to the embodiment of the present invention is explained with reference to FIG. **8**.

[0076] In step S**1**, the reservation assistance device **1** obtains the requested reservation data **11** for identifying the reservation contents requested by the customer. The requested reservation data **11** is used to identify the requested start time and the requested end time of the reservation as well as the requested time period that is the difference between the requested start time and the requested end time.

[0077] In step S**2**, the reservation assistance device **1** performs proposal processing by the proposal unit **22** for the requested reservation data **11** obtained in step S**1**, and obtains the new reservation data **14**.

[0078] In step S**3**, the reservation assistance device **1** updates the reservation data **15** according to the new reservation data **14** obtained in step S**2**, and also updates the status data **12**.

[0079] The proposal processing of step S**2** is explained with reference to FIG. **9**.

[0080] In step S**101**, the reservation assistance device **1** refers to the status data **12**, and checks that the reservation is acceptable in the period between the requested start time and requested end time. When the period between the requested start time and the requested end time includes the reservation unacceptable time period, the reservation assistance device **1** sends a request of changing the reservation contents to the customer terminal **2**.

[0081] In step S**102**, the reservation assistance device **1** determines whether or not the in-between time period is present before the requested start time. When the in-between time period is absent, the processing proceeds to step S**104**. When the in-between time period is present, in step S**103**, the reservation assistance device **1** sets the end time of the reservation unacceptable time period immediately before the requested start time, as the start candidate time, and proceeds to step S**104**. In the example illustrated in FIG. **2**, the reservation assistance device **1** sets the end time 14:00 of the reservation unacceptable time period immediately before the requested start time 14:30, as the start candidate time.

[0082] In step S**104**, the reservation assistance device **1** determines whether or not the in-between time period is present after the requested end time. When the in-between time period is absent, the processing proceeds to step S**106**. When the in-between time period is present, in step S**105**, the reservation assistance device **1** sets a time the requested time period before the start time of the reservation unacceptable time period immediately after the requested end time, as the start candidate time, and proceeds to step S**106**. In the example illustrated in FIG. **2**, the reservation assistance device **1** sets the time 15:00 the requested time period of one hour before the start time 16:00 of the reservation unacceptable time period immediately after the requested end time 15:30, as the start candidate time.

[0083] In step S**106**, the reservation assistance device **1** determines whether or not the start candidate time is set in step S**103** or step S**105**. When the start candidate time is not set, the processing proceeds to step S**110**. When the start candidate time is set, in step S**107**, the reservation assistance device **1** outputs the proposal of the change to the start candidate time, to the customer terminal **2**. When multiple times are set as the start candidate time, the reservation assistance device **1** sends each of the multiple times set as the start candidate time to the customer terminal **2**.

[0084] When the reservation assistance device **1** receives a reply of changing the requested start time from the customer terminal **2** in step S**108**, the processing proceeds to S**109**. When the reservation assistance device **1** receives a reply of not changing the requested start time, the processing proceeds to step S**110**.

[0085] In step S**109**, the reservation assistance device **1** generates the new reservation data **14** in which the start candidate time is set as the requested start time. When there are multiple start candidate times, the reservation assistance device **1** sets the start candidate time selected in the customer terminal, as the reservation start time.

[0086] In step S**110**, the reservation assistance device **1** generates the new reservation data **14** in which the requested start time is set as the requested start time.

[0087] When the new reservation data **14** is generated in step S**109** or step S**110**, the reservation assistance device **1** terminates the processing.

[0088] When the in-between time period is present before the requested start time requested by the customer or after the requested end time requested by the customer, the reservation assistance device **1** according to the embodiment of the present invention proposes a change of the reservation start time or the reservation end time to the customer to eliminate the in-between time period. When the instruction of accepting the proposal is inputted from the customer terminal **2**, the reservation assistance device **1** generates the new reservation data **14** in which the reservation start time is set such that the in-between time period is eliminated, and accepts the reservation. This can eliminate the in-between time period, and allows accepting of more reservations.

First Modified Example

[0089] Explanation is given of the case where the reservation assistance device **1** according to the embodiment of the present invention proposes the change of the reservation such that the in-between time period is eliminated, at the moment of accepting of reservation from the customer. Meanwhile, a reservation assistance device **1***a* according to a first modified example searches for the presence or absence of the in-between time period at an arbitrary timing after the accepting of reservation, and when the in-between time period is present, proposes a change of reservations before and after the in-between time period. The reservation assistance device **1***a* according to the first modified example may be combined with the reservation assistance device **1** according to the embodiment of the present invention to eliminate the in-between time period at the moment of reservation and at an arbitrary timing, or may not be combined with the reservation assistance device **1** according to the embodiment of the present invention to eliminate the in-between time period only at an arbitrary timing.

[0090] The reservation assistance device **1***a* according to the first modified example is explained with reference to FIG. **10**. The reservation assistance device **1***a* illustrated in FIG. **10** is different from the reservation assistance device **1** illustrated in FIG. **1** in that the reservation assistance device **1***a* includes no obtaining unit **21** and includes notification data **17** instead of the new reservation data **14**.

[0091] When a proposal unit **22***a* refers to the status data **12** and the difference time period between the first time period in which the reservation is unacceptable and the second time period in which the reservation is unacceptable and that is immediately after the first time period is equal to or more than the unit time period and is less than the predetermined time period, the proposal unit **22***a* proposes a change of the first time period or the second time period.

[0092] For example, assume that the reservation P**104** is fixed in the status display screen V**1** illustrated in FIG. **2**. In this case, the break P**102** of the staff is the first time period, and the reservation P**104** is the second time period. The proposal unit **22***a* determines whether or not the difference time period between the first time period and the second time period is equal to or more than the unit time period and is less than the predetermined time period. In the example illustrated in FIG. **2**, the difference time period is 30 minutes that is a difference between 14:00 being the end time of the first time period and 14:30 being the start time of the second time period. The difference time period of 30 minutes is equal to or more than the unit time period of 15 minutes, and is less than the predetermined time period of one hour. Accordingly, the proposal unit **22***a* determines that the in-between time period is present between the break P**102** that is the first time period and the reservation P**104** that is the second time period, and proposes the change of the first time period or the second time period to terminals of persons who have the right to change the respective time periods.

[0093] When the first time period is the change target and the reservation is unacceptable in the first time period due to the reservation of the customer, the proposal unit **22***a* outputs a proposal of changing the reservation end time to the start time of the second time period, to the customer terminal **2** of the customer who made the reservation for the first time period. When an instruction of accepting the proposal is inputted from the customer terminal **2**, the proposal unit **22***a* generates the notification data **17** for changing the reservation end time of the first time period to the start time of the second time period, and inputs the notification data **17** into the update unit **23**.

[0094] When the first time period is the change target and the reservation is unacceptable for the first time period due to the break of the staff, the proposal unit **22***a* outputs a proposal of changing the end time of the break to the start time of the second time period, to the staff terminal **3** of this staff. When an instruction of accepting the proposal is inputted from the staff terminal **3**, the proposal unit **22***a* generates the notification data **17** for changing the break end time of the first time period to the start time of the second time period, and inputs the notification data **17** into the update unit **23**. Although the case where the proposal is outputted to the staff terminal **3** is explained herein, the proposal may be outputted to an output device viewable by the staff such as an output device **906** of the reservation assistance device **1***a*.

[0095] Although explanation is given of the case where the first time period is postponed such that no in-between time period occurs between the first time period and the second time period, the present invention is not limited to this. For example, when the in-between time period is present between the first time period and the reservation unacceptable time period immediately before the first time period, the proposal unit **22***a* may propose a change of the start time of the first time period to the end time of the reservation unacceptable time period immediately before the first time period. When the instruction of accepting the proposal is inputted from the customer terminal **2** or the staff terminal **3**, the proposal unit **22***a* generates the notification data **17** for changing the start time of the first time period to the end time of the reservation unacceptable time period immediately before the first time period, and inputs the notification data **17** into the update unit **23**.

[0096] When the second time period is the change target and the reservation is unacceptable for the second time period due to the reservation of the customer, the proposal unit **22***a* outputs a proposal of changing the reservation start time to the end time of the first time period, to the customer terminal **2** of the customer who made the reservation for the second time period. When an instruction of accepting the proposal is inputted from the customer terminal **2**, the proposal unit

22*a* generates the notification data **17** for changing the reservation start time of the second time period to the end time of the first time period, and inputs the notification data **17** into the update unit **23**.

[0097] When the second time period is the change target and the reservation is unacceptable for the second time period due to the break of the staff, the proposal unit **22***a* outputs a proposal of changing the start time of the break to the end time of the first time period, to the staff terminal **3** of this staff. When an instruction of accepting the proposal is inputted from the staff terminal **3**, the proposal unit **22***a* generates the notification data **17** for changing the break start time of the second time period to the end time of the first time period, and inputs the notification data **17** into the update unit **23**.

[0098] Although explanation is given of the case where the second time period is advanced such that no in-between time period occurs between the first time period and the second time period, the present invention is not limited to this. For example, when the in-between time period is present between the second time period and the reservation unacceptable time period immediately after the second time period, the proposal unit **22***a* may propose a change of the end time of the second time period to the start time of the reservation unacceptable time period immediately after the second time period. When the instruction of accepting the proposal is inputted from the customer terminal **2** or the staff terminal **3**, the proposal unit **22***a* generates the notification data **17** for changing the end time of the second time period to the start time of the reservation unacceptable time period immediately after the second time period, and inputs the notification data **17** into the update unit **23**. The update unit **23** updates the reservation data **15**, the staff data **16**, and the status data **12** according to the notification data **17**.

[0099] In this case, the proposal unit **22***a* may propose a change of the reservation or the break in the order determined in advance for the first time period and the second time period.

[0100] For example, assume that one of the first time period and the second time period is the break hours of the staff, and the other is the reservation time period of the customer. First, the proposal unit **22***a* may proposes the change of the break hours of the staff, and when the proposal is not accepted, propose the change of the reservation of the customer.

[0101] The proposal unit **22***a* may determine a proposal destination such that no further proposal is made to a customer to which a proposal has been made in the past or a customer who did not accept the proposal. The proposal unit **22***a* holds the number of times of proposal, the number of times of unacceptance of proposal, the number of times of acceptance of proposal, and the like for each customer in the reservation data **15** or the status data **12**.

[0102] For example, when the proposal unit **22***a* has made the proposal predetermined times or more to a customer who has made the reservation for the first time period, the proposal unit **22***a* does not make the proposal to the customer of the first time period, and proposes a change of the second time period. Moreover, when the reservations for the first time period and the second time period are made, respectively, by different customers, the proposal unit **22***a* may preferentially propose the change of the reservation to a customer selected from the different customers based on a predetermined standard such as a customer to which the number of times of proposal made in the past is fewer, a customer who has rejected the proposal fewer times in the past, or a customer who has accepted the proposal more times in the past.

[0103] Moreover, when the proposal unit **22***a* refers to the status data **12** for one day and finds multiple in-between time periods, the proposal unit **22***a* may identify the reservation unacceptable time period that can efficiently eliminate the in-between time periods, and propose a change to the identified reservation unacceptable time period.

[0104] For example, in the example illustrated in FIG. **2**, when the reservation P**104** is fixed, the in-between time periods are present between the break **102** and the reservation P**104** and between the reservation P**104** and the reservation P**103**. In this case, (i) a time change of the break P**102** and the reservation P**103** or (ii) a time change of the reservation P**104** can eliminate two in-between time periods. In (i), postponing the break P**102** by 30 minutes and forwarding the reservation P**103** by 30 minutes can make the in-between time periods before and after the reservation P**104** zero. Alternatively, forwarding the break P**102** by 30 minutes and postponing the reservation P**103** by 30 minutes can create free time periods of one hour before and after the reservation P**104**. In (ii), forwarding the reservation P**104** by 30 minutes can make the in-between time period before the reservation P**104** zero, and create a free time period of one hour after the reservation P**104**. Alternatively, postponing the reservation P**104** by 30 minutes can make the in-between time period after the reservation P**104** zero, and create a free time period of one hour before the reservation P**104**.

[0105] In the plan (i), the in-between time periods are not eliminated unless the change is accepted for both of the break P**102** and the reservation P**103**. Meanwhile, in the plan (ii), the in-between time periods are eliminated only with the change of the reservation P**104**. Accordingly, the proposal unit **22***a* may propose the reservation change according to (ii) with fewer change proposal destinations.

[0106] The proposal processing by the proposal unit **22***a* according to the first modified example is explained with reference to FIG. **11**. In the example of FIG. **11**, explanation is given of the case where a change of the reservation unacceptable time period due to the staff is preferentially proposed.

[0107] In step S**201**, the proposal unit **22***a* refers to the status data **12**, and identifies the first time period in which the reservation is unacceptable and the second time period in which the reservation is unacceptable and that is immediately after the first the time period. In step S**202**, the proposal unit **22***a* determines whether the difference time period between the first time period and the second time period identified in step S**201** is the in-between time period or not. When the difference time period is not the in-between time period, the proposal unit **22***a* terminates the processing.

[0108] When the difference time period is the in-between time period, in step S**203**, the proposal unit **22***a* determines whether or not the reservation unacceptable time period due to the staff is present in the first time period and the second time period. When the reservation unacceptable time period due to the staff is absent, the processing proceeds to step S**206**. When the reservation unacceptable time period due to the staff is present, in step S**204**, the proposal unit **22***a* outputs a proposal of changing the reservation unacceptable time period due to the staff, to the staff terminal **3**. When an instruction of no change is inputted from the staff terminal **3**, the processing proceeds to step S**206**. When an instruction of change is inputted, the processing proceeds to step S**209**.

[0109] In step S**206**, the proposal unit **22***a* outputs a proposal of changing the reservation unacceptable time period to the customer terminal **2** of one of the customers of the first time period and the second time period or each of the customers. When the instruction of change is inputted, the processing proceeds to step S**209**. When the instruction of no change is inputted from the staff terminal **3**, the processing is terminated.

[0110] In step S**209**, the proposal unit **22***a* generates the notification data **17** for changing the reservation unacceptable time period for which the staff or the customer has accepted the proposal of change. The proposal unit **22***a* inputs the notification data **17** into the update unit **23**, and terminates the processing.

[0111] The reservation assistance device **1***a* according to the first modified example searches for the in-between time period at an arbitrary timing even after the reservation of the customer, and proposes the change of the already-made reservation of the customer or the plan of the staff such that the in-between time period is eliminated. This can eliminate the in-between time period, and allows accepting of more reservations. Moreover, even if the customer does not accept the change of reservation at the moment of the reservation in the embodiment of the present invention, the already-made reservation can be changed. Accordingly, the in-between time period can be eliminated more efficiently.

[0112] When the reservation assistance device **1***a* of the first modified example requests the change of reservation, the reservation assistance device **1***a* may propose the change of the reservation start time such that no in-between time period occurs before and after the reservation. Alternatively, the reservation assistance device **1***a* may propose the change of the reservation start time such that the in-between time period does not occur in at least one of periods before and after the reservation or is reduced. The "in-between time period does not occur" means that, when the in-between time period of 30 minutes is present before the reservation start time, the in-between time period is made zero by forwarding the reservation start time by 30 minutes or a free time period of one hour or more is created by postponing the reservation start time by 30 minutes or more. The "in-between time period is reduced" means that, when the in-between time period of 30 minutes is present before the reservation start time, the in-between time period is reduced from 30 minutes to 15 minutes by forwarding the reservation start time by 15 minutes.

Second Modified Example

[0113] In the embodiment and the first modified example of the present invention, explanation is given of the case where the reservation is changed when the in-between time period occurs at the moment of accepting of reservation or after accepting of reservation. Meanwhile, a reservation assistance device **1***b* according to a second modified example performs control such that no in-between time period occurs by not including time periods that cause the in-between time period in candidates of the start time of reservation when the customer specifies the start time of reservation.

[0114] The reservation assistance device **1***b* according to the second modified example is explained with reference to FIG. **12**. The reservation assistance device **1***b* according to the second modified example is different from the reservation assistance device **1** according to the embodiment of the present invention in the obtaining unit **21** and the contents of the requested reservation data **11**, and the reservation assistance device **1***b* includes start candidate time data **18**, an identification unit **24**, and a display unit **25** instead of the new reservation data **14** and the proposal unit **22**. The status data **12** and the setting data **13** in FIG. **12** are as explained in the embodiment of the present invention and the like. Note that, although omitted in the example illustrated in FIG. **12**, the reservation assistance device **1***b* according to the second modified example may also include the reservation data **15**, the staff data **16**, and the update unit **23** illustrated in FIG. **1** like the reservation assistance device **1** according to the embodiment of the present invention.

[0115] The obtaining unit **21** obtains the requested time period of the reservation requested by the customer. The obtaining unit **21** obtains the requested time period of the reservation itself from the customer terminal **2**. Alternatively, the obtaining unit **21** obtains a requested practice menu from the customer terminal **2**, and obtains the requested time period of the reservation based on a time period for providing the obtained practice menu. The obtaining unit **21** may obtain a requested date on which the reservation is requested, in addition to the requested time period.

[0116] The identification unit **24** identifies the start candidate times of the reservation at which the reservation is acceptable for the requested time period of the reservation and at which no in-between time period occurs in a period before or after the reservation. Specifically, the identification unit **24** identifies the start candidate times of the reservation by excluding the start candidate times of the reservation at which the in-between time period occurs in both of the periods before and after the reservation from the start candidate times of the reservation at which the reservation is acceptable for the requested time period of the reservation. In the case where the obtaining unit **21** also obtains the requested date, the identification unit **24** identifies the start candidate times in the requested date. Although the case where the start candidate times of the reservation at which no in-between time period occurs in at least one of the periods before and after the reservation is at least identified is explained in the second modified example, the start candidate times of the reservation at which no in-between time period occurs in both of the periods before and after the reservation may be identified.

[0117] The identification unit **24** refers to the status data **12**, and identifies the start candidate time at which the difference time period between the start candidate time and the end time of the reservation unacceptable time period immediately before the start candidate time is zero or equal to or more than the predetermined time period or the start candidate time at which the difference time period between a time at which the requested time period has elapsed from the start candidate time to the start time of the reservation unacceptable time period immediately after this time is zero or equal to or more than the predetermined time period, from the start candidate times at which the reservation of the requested time period is acceptable.

[0118] The display unit **25** displays the start candidate times identified by the identification unit **24**, on the customer terminal **2**. The reservation assistance device **1**_b_ receives the start time specified from the start candidate times by the customer from the customer terminal **2**, and generates the new reservation data **14** in which: the reservation start time is set to the start time specified by the customer; the reservation end time is set to a time at which the requested time period has elasped from the start time specified by the customer; and the reservation time period is set to the requested time period. The reservation assistance device **1**_b_ updates the status data **12** and the reservation data **15** according to the new reservation data **14**.

[0119] Explanation is given of processing in the case where the identification unit **24** identifies the start candidate times of the reservation by excluding the start candidate times of the reservation at which the in-between time period occurs in both of the periods before and after the reservation in the second modified example. The identification unit **24** identifies the start candidate times of the reservation by allowing the case where no in-between time period occurs in both of the periods before and after the reservation, the case where the in-between time period occurs in the period before the reservation but not in the period after the reservation, and the case where the in-between time period occurs in the period after the reservation but not in the period before the reservation.

[0120] Explanation is given of processing of the identification unit **24** in the case where the reservation P**101**, the break P**102**, and the reservation P**103** are fixed in the example illustrated in FIG. **2** and the customer specifies one hour as the requested time period.

[0121] In the example illustrated in FIG. **2**, the start candidate times at which the reservation for the time period of one hour requested by the customer is acceptable are 10:30, 10:45, 11:00, 11:15, 11:30, 11:45, 12:00, 14:00, 14:15, 14:30, 14:45, and 15:00.

[0122] When the start candidate time is set to 10:30, the difference time period between the start candidate time 10:30 and the end time 10:30 of the reservation unacceptable time period immediately before the start candidate time is zero, and no in-between time period occurs. When the start candidate time is set to 10:45, the difference time period between the start candidate time 10:45 and the end time 10:30 of the reservation unacceptable time period immediately before this time 10:45 is 15 minutes, and the in-between time period thus occurs. Meanwhile, since the difference time period between the time 11:45 at which the requested time period of one hour has elapsed from the start candidate time 10:45 and the start time 13:00 of the reservation unacceptable time period immediately after this time 11:45 is one hour or more, no in-between time period occurs. Similarly, when the start candidate time is set to 11:00, 11:30, 11:45, 12:00, 14:00, or 15:00, no in-between time period occurs in at least one of the period before the start candidate time and the period after the time at which the requested time period of one hour has elapsed from the start candidate time. Accordingly, the start candidate times are 10:30, 10:45, 11:00, 11:30, 11:45, 12:00, 14:00, and 15:00.

[0123] Meanwhile, when the start candidate time is set to 11:15, the difference time period between the start candidate time 11:15 and the end time 10:30 of the reservation unacceptable time period immediately before the start candidate time is 45 minutes, and the in-between time period occurs. Moreover, since the difference time period between 12:15 that is one hour after the start candidate time 11:15 and the start time 13:00 of the reservation unacceptable time period immediately after this time 12:15 is 45 minutes, the in-between time period occurs. When the start candidate time is set to 11:15, the in-between time period occurs in both of the period before the start candidate time

and the period after the time at which the requested time period of one hour has elapsed from the start candidate time. Similarly, when the start candidate time is set to 14:15, 14:30, or 14:45, the in-between time period occurs in both of the period before the start candidate time and the period after the time at which the requested time period of one hour has elasped from the start candidate time. Accordingly, 11:15, 14:15, 14:30, and 14:45 are excluded from the start candidate times.
[0124] Accordingly, in the case of the example illustrated in FIG. **2**, the identification unit **24** identifies the start times 10:30, 10:45, 11:00, 11:30, 11:45, 12:00, 14:00, and 15:00 at which no in-between time period occurs in at least one of the period before the start time and the period after the end time, as the start candidate times, from among the start times 10:30, 10:45, 11:00, 11:15, 11:30, 11:45, 12:00, 14:00, 14:15, 14:30, 14:45, and 15:00 at which the reservation for the time period of one hour requested by the customer is acceptable.
[0125] As illustrated in FIG. **13**, the display unit **25** displays a selection screen V**4** in which one of the start candidate times identified by the identification unit **24** can be selected, on the customer terminal **2**, and causes the customer to select one of the start candidate times.
[0126] The processing of the identification unit **24** is an example, and the present invention is not limited to this. For example, if the reservation assistance device **1***b* accepts only the reservation for which no in-between time period occurs in one of the periods before and after the reservation time period in a predetermined condition such as the case where the number of reservation in one day reaches or exceeds a predetermined number or the case where the number of days between the current date and the date of reservation is small, the reservation assistance device **1***b* may be unable to accept the reservation itself. Accordingly, when the predetermined condition that allows the in-between time period is satisfied, the identification unit **24** may change the identification method of the start candidate times such that more start candidate times are identified. For example, the identification unit **24** may identify not only the start times at which no in-between time period occurs in at least one of the periods before and after the reservation time period, but also the start times at which the in-between time period occurs in both of the periods before and after the reservation time period, as the start candidate times.
[0127] Moreover, in a predetermined condition such as the case where the number of days between the current date and the date of the reservation is large, the reservation assistance device **1***b* may accept only the reservation for which no in-between time period occurs in both of the periods before and after the reservation time period. When the predetermined condition that does not allow the in-between time period is satisfied, the identification unit **24** may identify the start candidate times at which no in-between time period occurs in both of the periods before and after the reservation time period.
[0128] Explanation is given of the case where the reservation assistance device **1***b* accepts only the reservation for which no in-between time period occurs in both of the periods before and after the reservation time period. In this case, the identification unit **24** identifies only the case where no in-between time period occurs in both of periods before and after the reservation, as the start candidate time. The case where the in-between time period occurs in the period before the reservation but not in the period after the reservation and the case where the in-between time period occurs in the period after the reservation but not in the period before the reservation are excluded from the start candidate times.
[0129] In the example illustrated in FIG. **2**, the start candidate times at which the reservation for the time period of one hour requested by the customer is acceptable are 10:30, 10:45, 11:00, 11:15, 11:30, 11:45, 12:00, 14:00, 14:15, 14:30, 14:45, and 15:00. When 10:45 and 11:00 among these start candidate times are each set as the start candidate time, no in-between time period occurs in the period after the time at which the requested time period of one hour has elapsed the start candidate time, but the in-between time period occurs in the period before the start candidate time. When 11:30 and 11:45 are each set as the start candidate time, no in-between time period occurs in the period before the start candidate time, but the in-between time period occurs in the period after

the time at which the requested time period of one hour has elapsed from the start candidate time. When 11:15, 14:15, 14:30, and 14:45 are each set as the start candidate time, the in-between time period occurs in both of the period before the start candidate time and the period after the time at which the requested time period of one hour has elapsed from the start candidate time.

[0130] Accordingly, when the reservation assistance device **1***b* accepts only the reservation for which no in-between time period occurs in both of the periods before and after the reservation time period, the identification unit **24** identifies 10:30, 12:00, 14:00, and 15:00 as the start candidate times.

[0131] Processing of the reservation assistance device **1***b* according to the second modified example is explained with reference to FIG. **14**. Explanation is given of processing in the case where reservation assistance device **1***b* identifies the start candidate times of the reservation by excluding the start candidate times of the reservation at which the in-between time period occurs in at least one of the periods before and after the reservation.

[0132] In step S**51**, the reservation assistance device **1***b* obtains the requested time period and the requested date for which the customer requests to make the reservation.

[0133] The reservation assistance device **1***b* performs processing of step S**52** to step S**56** for each of target times of the requested date. The target times are set every unit time period.

[0134] In step S**52**, the reservation assistance device **1***b* refers to the status data **12**, and determines whether or not the reservation is acceptable in a period of the requested time period from the target time. When the period includes the reservation unacceptable time period, the processing proceeds to step S**56**. When the reservation is acceptable, the processing proceeds to step S**53**.

[0135] In step S**53**, the reservation assistance device **1***b* determines whether or not the in-between time period occurs before the target time. When no in-between time period occurs, the processing proceeds to step S**55**. Meanwhile, when the in-between time period occurs before the target time, the processing proceeds to step S**54**.

[0136] In step S**54**, the reservation assistance device **1***b* determines whether or not the in-between time period occurs after elapse of the requested time period from the target time. When the in-between time period occurs, the processing proceeds to step S**56**. When no in-between time period occurs, the processing proceeds to step S**55**.

[0137] In step S**55**, the reservation assistance device **1***b* sets the target time as the start candidate time. Meanwhile, in step S**56**, the reservation assistance device **1***b* excludes the target time from the start candidate times.

[0138] When the processing from step S**52** to step S**56** is completed for each target time, the processing proceeds to step S**57**. In step S**57**, the reservation assistance device **1***b* displays each of the start candidate times set in step S**55** on the customer terminal **2**. The reservation assistance device **1***b* generates the new reservation data **14** with the start candidate time selected in the customer terminal **2** set as the reservation start time.

[0139] The reservation assistance device **1***b* according to the second modified example displays only the start candidate times of the reservation at which no in-between time period occurs in at least one of the periods before and after the reservation, on the customer terminal **2**, and causes the customer to select the start time requested by the customer from the start candidate times. The reservation assistance device **1***b* can thereby assist streamlining of businesses.

[0140] After the reservation is made by the second modified example by identifying the start time of the reservation from the start candidate times of the reservation at which no in-between time period occurs in at least one of the periods before and after the reservation, this reservation or the reservation unacceptable time period before or after this reservation may be shifted as explained in the first modified example. The reservation assistance device **1***b* can thereby eliminate the in-between time period.

Third Modified Example

[0141] In the embodiment of the present invention, there is a case where no in-between time period

occurs before the reservation start time or after the reservation end time at the moment of reservation by the customer. However, there is a case where, after this reservation of the customer, the in-between time period occurs afterwards due to accepting of a reservation of another customer. In the third modified example, elimination of the in-between time period is made possible also in the case where the in-between time period occurs afterwards.
[0142] Accordingly, in the third modified example, when no in-between time period occurs before the reservation start time and after the reservation end time at the moment of reservation by the customer, the reservation assistance device **1** inquires the customer whether or not the reservation can be changed afterwards. When the customer answers that the reservation change is possible, the reservation assistance device **1** sets this information in the new reservation data **14**, and reflects the information in the reservation data **15**.
[0143] For example, when it is before the execution of step S**3** in FIG. **8** and no in-between time period occurs before the reservation start time requested by the customer and after the reservation end time requested by the customer, the reservation assistance device **1** displays a selection screen V**5** illustrated in FIG. **15**, and inquires the customer whether or not the reservation can be changed afterwards. Alternatively, when it is before the execution of the processing of step S**110** in FIG. **9** and the customer does not agree with a change to one of the start time candidates presented by the reservation assistance device **1**, the reservation assistance device **1** may display the selection screen V**5** illustrated in FIG. **15**, and inquire the customer whether or not the reservation can be changed afterwards. In this case, the inquiry of step S**107** in FIG. **9** and the inquiry of the selection screen V**5** may be implemented on one screen. The selection screen V**5** causes the customer to input whether or not advancing or delaying of the start time is possible and, when the change is possible, a time period in which the change is allowable. In step S**110**, the reservation assistance device **1** sets the answer inputted in the selection screen V**5** in the new reservation data **14**, and reflects the answer in the reservation data **15**.
[0144] In the third modified example, the reservation assistance device **1** refers to the status data **12** at a predetermined timing, and when the difference time period between the first time period in which the reservation is unacceptable and the second time period in which the reservation is unacceptable and that is immediately after the first time period is equal to or larger than the unit time period and is less than the predetermined time period, propose a change of the first time period or the second time period. This timing is, for example, a timing at which no change of the reservation occurs such as after an end of a period of accepting a reservation change or cancel. Alternatively, this timing is a timing at which a reservation change is less likely to occur such after business close of a day before the reservation date. As another alternative, this timing is a timing at which occurrence of in-between time period due to accepting of a new reservation is found.
[0145] In this case, when the customer who made the reservation for the first time period is the customer who answered that the time change is possible at the moment of reservation, the reservation assistance device **1** outputs a proposal of changing the reservation end time to the start time of the second time period, to this customer. When the customer who made the reservation for the second time period is the customer who answered that the time change is possible at the moment of reservation, the reservation assistance device **1** outputs a proposal of changing the reservation start time to the end time of the first time period, to this customer. This proposal is notified by an e-mail, an SMS (short message service), an application, an SNS (social networking service) or the like. The reply of the customer to the proposal may be sent back by means by which the reservation assistance device **1** has made the proposal, or by a web site to which the customer is guided by a URL described in notification contents. The proposal and the reply can be made by any means as long as the proposal reaches the customer from the reservation assistance device **1** and the reply of the customer to this proposal reaches the reservation assistance device **1**.
[0146] When the difference time period between the first time period and the second time period is reduced by time change contents allowed by the customer in advance, the reservation assistance

device **1** may propose the time change. For example, when the customer is a person who made the reservation for the first time period and who set that the reservation can be postponed by 30 minutes and the difference time period between the first time period and the second time period is 45 minutes, postponing the reservation by the customer can reduce the difference time period from 45 minutes to 15 minutes. The reservation assistance device **1** proposes postponing of the reservation by 30 minutes to the customer. When the difference time period between the first time period and the second time period becomes equal to or more than the predetermined time period by the time change contents allowed by the customer in advance, the reservation assistance device **1** may propose the time change. For example, when the customer is a person who made the reservation for the first time period and who set that the reservation can be forwarded by 30 minutes and the difference time period between the first time period and the second time period is 45 minutes, advancing the reservation by the customer can increase the difference time period from 45 minutes to one hour and 15 minutes, and accepting of a new reservation becomes possible. When the difference time period between the first time period and the second time period becomes zero by the time change contents allowed by the customer in advance, the reservation assistance device **1** may propose the time change.

[0147] Note that, although the case where the customer sets whether or not the time change is possible at the moment of reservation is explained in the third modified example, the present invention is not limited to this. When the customer performs default setting or the like and answers in advance that the time change is possible before the finding of occurrence of the in-between time period before or after the reservation at any of the moment of reservation, the moment before the reservation, and the moment after the reservation of the customer, the reservation assistance device **1** proposes the time change to this customer.

[0148] When the customer accepts the proposal by the reservation assistance device **1**, the in-between time period can be reduced. This allows the staff to deal with a new customer or have solid blocks of time for work. This allows effective use of work time and an increase of sales in the shop.

Fourth Modified Example

[0149] Explanation is given of the case where the reservation assistance device **1** according to a fourth modified example performs control such that, when the customer specifies the start time of the reservation, the time at which the in-between time period occurs is made less likely to be selected as the candidate of the start time, and the in-between time period is thereby made less likely to occur. In comparison to the second modified example, in the fourth modified example, the reservation assistance device **1** allows the customer to select the time at which the in-between time period occurs, but displays the candidates of the start time such that the time at which no in-between time period occurs is more likely to be selected.

[0150] Specifically, the display unit **25** of the reservation assistance device **1** displays a selection screen V**6** illustrated in FIG. **16**A on the customer terminal **2** to allow selection of the reservation start time. The selection screen V**6** displays the start candidate times at which no in-between time period occurs in at least one of the periods before and after the reservation time period, among the start candidate times at which the reservation of the requested time period specified by the customer is acceptable, in a list box. Specifically, each of the start candidate times at which no in-between time period occurs in the periods before and after the reservation time period is such a time that the difference time period between the start candidate time and the end time of the reservation unacceptable time period immediately before the start candidate time is zero or equal to or more than the predetermined time, or the difference time period between the time at which the requested time period has elapsed from the start candidate time and the start time of the reservation unacceptable time period immediately after this time is zero or equal to or more than the predetermined time period.

[0151] As illustrated in FIG. **16**A, the display unit **25** displays the start candidate times at which no in-between time period occurs in the periods before and after the reservation time period, in the list box of the selection screen V**6** for selecting the reservation start time. The start candidate times at which no in-between time period occurs in at least one of the periods before and after the reservation time period are 10:30, 10:45, 11:00, 11:30, 11:45, 12:00, 14:00, and 15:00 as explained in the second modified example.

[0152] The identification unit **24** of the reservation assistance device **1** identifies, in addition to the start candidate times at which no in-between time period occurs in at least one of the periods before and after the reservation time period, sub-start candidate times at which the in-between time period occurs in both of periods before and after the reservation time period. The sub-start candidate times are each the start candidate time at which the difference time period between the start candidate time and the end time of the reservation unacceptable time period immediately before the start candidate time is less than the predetermined time period and the difference time period between the time at which the requested time period has elapsed from the start candidate time and the start time of the reservation unacceptable time period immediately after this time is less than the predetermined time period, among the start candidate times at which the reservation of the requested time period specified by the customer is acceptable.

[0153] As illustrated in FIG. **16**B, the display unit **25** displays the start candidate times at which no in-between time period occurs in the periods before and after the reservation time period in the list box of a selection screen V**6***a* in response to an instruction of the customer, and when a development instruction portion P**501** provided between 11:00 and 11:30 is selected in the selection screen V**6** of FIG. **16**A, displays the sub-start candidate time in the selection screen V**6***a* illustrated in FIG. **16**B. In FIG. **16**B, 11:15 is displayed as the sub-start candidate time between the start candidate times 11:00 and 11:30. The customer can thereby select not only the start candidate times at which no in-between time period occurs, but also the sub-start candidate times at which the in-between time period occurs.

[0154] Specifically, when the sub-start candidate time is present between two adjacent start candidate times among the start candidate times, the display unit **25** displays the development instruction portion corresponding to the two adjacent start candidate times in the list box. When the development instruction portion is selected, the display unit **25** displays the sub-start candidate time between the two adjacent start candidate times in the list box.

[0155] In the selection screen V**6** of FIG. **16**A, the development instruction portion P**501** is displayed between the start candidate times 11:00 and 11:30. A development instruction portion P**502** is displayed between the start candidate times 12:00 and 14:00. A development instruction portion P**503** is displayed between the start candidate times 14:00 and 15:00.

[0156] When the customer clicks the development instruction portion P**501** in this case, the display unit **25** displays 11:15 as illustrated in FIG. **16**B as the sub-start candidate time corresponding to the development instruction portion **501**.

[0157] Similarly, when the customer clicks the development instruction portion P**502**, the display unit **25** displays times in increments of the unit time period from 12:15 to 13:45, as the sub-start candidate times corresponding to the development instruction portion P**502**. When the customer clicks the development instruction portion P**503**, the display unit **25** displays times in increments of the unit time period from 14:15 to 14:45, as the sub-start candidate times corresponding to the development instruction portion P**503**. Note that the unit time period is 15 minutes in the present disclosure.

[0158] In the selection screen V**6***a* of FIG. **16**B, collapse instruction portions P**501***a* and P**501***b* are displayed to correspond to the portions where the sub-start candidate times are displayed. When the collapse instruction portion P**501***a* or P**501***b* is selected, the display unit **25** displays the selection screen V**6** illustrated in FIG. **16**A. The display unit **25** cancels display of the sub-start candidate times and the collapse instruction portions P**501***a* and P**501***b*, and displays the development instruction portion P**501** in the selection screen V**6**.

[0159] Although the development instruction portion P**501** and the like are icons in which an upward arrow and a downward arrow are vertically arranged in FIGS. **16**A and B, the present invention is not limited to this. Although explanation is given of the case where the collapse instruction portion P**501***a* is an icon including a downward arrow while the collapse instruction portion P**501***b* is an icon including an upward arrow and these icons are displayed, respectively, above and below a display portion of the sub-start candidate times, the present invention is not limited to this. Moreover, although the case where the start candidate times and the sub-start candidate times are displayed in the list box is explained in FIGS. **16**A and B, the present invention is not limited to this. The start candidate times and the sub-start candidate times may be displayed in any way as long as only the start candidate times or the start candidate times and the sub-start candidate times are displayed in response to an instruction of the customer, and one of the times is identified as the reservation start time.

[0160] Processing of the reservation assistance device **1** according to the fourth modified example is explained with reference to FIGS. **17** to **18**.

[0161] In step S**71**, the reservation assistance device **1** obtains the requested time period and the requested date for which the customer requests to make the reservation.

[0162] The reservation assistance device **1** performs processing of step S**72** to step S**76** for each of target times of the requested date. In this case, the target times are set every unit time period.

[0163] In step S**72**, the reservation assistance device **1** refers to the status data **12**, and determines whether or not the reservation is acceptable in a period of the requested time period from the target time. When the period includes the reservation unacceptable time period, the reservation assistance device **1** performs the processing of S**72** for the next target time. When the reservation is acceptable, the processing proceeds to step S**73**.

[0164] In step S**73**, the reservation assistance device **1** determines whether or not the in-between time period occurs before the target time. When the in-between time period occurs, the processing proceeds to step S**74**. Meanwhile, when no in-between time period occurs before the target time, the processing proceeds to step S**75**.

[0165] In step S**74**, the reservation assistance device **1** determines whether or not the in-between time period occurs after elapse of the requested time period from the target time. When no in-between time period occurs, the processing proceeds to step S**75**. When the in-between time period occurs, the processing proceeds to step S**76**.

[0166] In step S**75**, the reservation assistance device **1** determines that no in-between time period occurs in at least one of the periods before and after the target time, and sets the target time as the start candidate time. Meanwhile, in step S**76**, the reservation assistance device **1** determines that the in-between time period occurs in both of the periods before and after the target time, and sets the target time as the sub-start candidate time.

[0167] When the processing from step S**72** to S**76** is completed for each target time, the processing proceeds to step S**81**. Note that, when the reservation is unacceptable for all target times, the reservation assistance device **1** displays this information on the customer terminal **2**.

[0168] In step S**81**, the reservation assistance device **1** first displays the start candidate times identified in step S**75** on the customer terminal **2**. In step S**82**, the reservation assistance device **1** determines whether or not any of the sub-start candidate times identified in step S**76** is present between the adjacent start time candidates. When the sub-start time candidate is present, in step S**83**, the reservation assistance device **1** displays the development instruction portion on the customer terminal **2** in association with the adjacent start candidate times.

[0169] In step S**84**, when an instruction inputted from the customer terminal **2** is selection of the development instruction portion, the processing proceeds to step S**85**. In step S**85**, the reservation assistance device **1** displays the sub-start candidate time between the two start candidate times corresponding to the selected development instruction portion and the collapse instruction portion for hiding the sub-start candidate time, on the customer terminal **2**.

[0170] When the instruction inputted from the customer terminal **2** is selection of the collapse instruction portion in step S**84**, the processing proceeds to step S**86**. In step S**86**, the reservation assistance device **1** hides the sub-start candidate time corresponding to the selected collapse instruction portion, and displays the development instruction portion.

[0171] When the instruction inputted from the customer terminal **2** is selection of the time in step S**84**, the processing proceeds to step S**87**. In step S**87**, the reservation assistance device **1** generates the new reservation data **14** with the selected time set as the reservation start time.

[0172] In the fourth modified example, the reservation assistance device **1** first displays the start candidate times at which no in-between time period occurs in at least one of the periods before and after the reservation time period as the reservation acceptable times, and also displays the icon for displaying the sub-start candidate times at which the in-between time period occurs in the periods before and after the reservation time period, on the customer terminal **2**. Moreover, the reservation assistance device **1** displays the sub-start candidate times in addition to the start candidate times in response to selection of this icon.

[0173] The reservation assistance device **1** displays the sub-start candidate times only after an action performed by the customer. The reservation assistance device **1** can promote the customer to select the start candidate times displayed first. The reservation assistance device **1** can control the start time of the reservation such that no in-between time period occurs.

[0174] For example, a general-purpose computer system including the CPU (central processing unit, processor) **901**, the memory **902**, the storage **903** (HDD: hard disk drive, SSD: solid state drive), a communication device **904**, an input device **905**, and the output device **906** is used as the reservation assistance device **1** (including the reservation assistance devices **1***a* and **1***b*) of the present embodiment explained above. In this computer system, the CPU **901** executes a program loaded onto the memory **902** to implement the functions of the reservation assistance device **1**.

[0175] Note that the reservation assistance device **1** may be implemented by one computer or by multiple computers. Moreover, the reservation assistance device **1** may be a virtual machine implemented in a computer.

[0176] The program of the reservation assistance device **1** may be stored in a computer-readable recording medium such as an HDD, an SSD, a USB (Universal Serial Bus) a memory, a CD (Compact Disc), or a DVD (Digital Versatile Disc) or may be distributed through a network. The computer-readable recording medium is, for example, a non-transitory recording medium.

[0177] Note that the present invention is not limited to the above embodiment, and various modifications can be made within the scope of the spirit of present invention.

EXPLANATION OF THE REFERENCE NUMERALS

[0178] **1** reservation assistance device [0179] **2** customer terminal [0180] **3** staff terminal [0181] **5** processing system [0182] **11** requested reservation data [0183] **12** status data [0184] **13** setting data [0185] **14** new reservation data [0186] **15** reservation data [0187] **16** staff data [0188] **17** notification data [0189] **18** start candidate time data [0190] **21** obtaining unit [0191] **22** proposal unit [0192] **23** update unit [0193] **24** identification unit [0194] **25** display unit [0195] **901** CPU [0196] **902** memory [0197] **903** storage [0198] **904** communication device [0199] **905** input device [0200] **906** output device

Claims

1. A reservation assistance device comprising: a storage device configured to store status data indicating whether or not a reservation from a customer is acceptable for each of unit time periods; an obtaining unit, including one or more processors, configured to obtain a requested start time and a requested end time of the reservation requested by the customer; and a proposal unit, including one or more processors, configured to refer to the status data, and when the reservation is acceptable in a period from the requested start time to the requested end time and a difference time period between the requested start time and an end time of a reservation unacceptable time period

immediately before the requested start time is equal to or larger than a unit time period and is less than a predetermined time period, output a proposal of changing the requested start time to the end time of the reservation unacceptable time period immediately before the requested start time, to the customer.
2. The reservation assistance device according to claim 1, wherein, when a difference time period between the requested end time and a start time of the reservation unacceptable time period immediately after the requested end time is equal to or larger than the unit time period and is less than the predetermined time period, the proposal unit outputs a proposal of changing the requested end time to the start time of the reservation unacceptable time period immediately after the requested end time.
3. A reservation assistance device comprising: a storage device configured to store status data indicating whether or not a reservation from a customer is acceptable for each of unit time periods; and a proposal unit, including one or more processors, configured to refer to the status data, and when a difference time period between a first time period in which the reservation is unacceptable and a second time period in which the reservation is unacceptable and that is immediately after the first time period is equal to or larger than a unit time period and is less than a predetermined time period, propose a change of the first time period or the second time period.
4. The reservation assistance device according to claim 3, wherein, when the reservation is unacceptable in the first time period due to the reservation of the customer, the proposal unit outputs a proposal of changing a reservation end time to a start time of the second time period, to the customer.
5. The reservation assistance device according to claim 3, wherein, when the reservation is unacceptable in the first time period due to a break of a staff, the proposal unit outputs a proposal of changing an end time of the break to a start time of the second time period, to the staff.
6. The reservation assistance device according to claim 3, wherein, when the proposal unit has made the proposal a predetermined number of times or more to the customer who has made the reservation for the first time period, the proposal unit proposes a change of the second time period.
7. The reservation assistance device according to claim 4, wherein the customer is a customer who has answered in advance that a time change is possible.
8. The reservation assistance device according to claim 7, wherein, when the difference time period is reduced or is made to be equal to or more than the predetermined time period by time change contents allowed by the customer in advance, the proposal unit makes the proposal.
9. A reservation assistance device comprising: a storage device configured to store status data indicating whether or not a reservation from a customer is acceptable for each of unit time periods; an obtaining unit, including one or more processors, configured to obtain a requested time period of the reservation requested by the customer; and an identification unit, including one or more processors, configured to refer to the status data and identify, from among start candidate times at which the reservation of the requested time period is acceptable, a start candidate time at which a difference time period between the start candidate time and an end time of a reservation unacceptable time period immediately before the start candidate time is zero or equal to or more than a predetermined time period or a difference time period between a time at which the requested time period has elapsed from the start candidate time and a start time of the reservation unacceptable time period immediately after the time is zero or equal to or more than the predetermined time period; and a display unit configured to display the start candidate time identified by the identification unit, to the customer.
10. The reservation assistance device according to claim 9, wherein the identification unit further identifies, from among the start candidate times at which the reservation of the requested time period is acceptable, the start candidate time at which the difference time period between the start candidate time and the end time of the reservation unacceptable time period immediately before the start candidate time is less than the predetermined time period and the difference time period

between the time at which the requested time period has elapsed from the start candidate time and the start time of the reservation unacceptable time period immediately after the time is less than the predetermined time period, as a sub-start candidate time, and the display unit further displays the sub-start candidate time in response to an instruction of the customer.

11. The reservation assistance device according to claim 10, wherein when the sub-start candidate time is present between two adjacent start candidate times among the start candidate times, the display unit displays a development instruction portion such that the development instruction portion corresponds to the two adjacent start candidate times, and when the development instruction portion is selected, the display unit displays the sub-start candidate time between the two adjacent start candidate times.

12. (canceled)

13. (canceled)

14. (canceled)

15. (canceled)